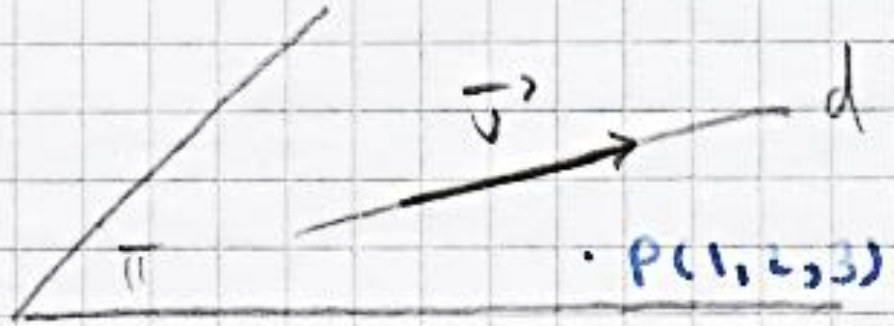
MATEMATICI 2

Seminar 4

Probl. Să se det. simetricul lui $P(1,2,3)$ în rap. cu dr. $d: \frac{x-1}{7} = \frac{y+1}{5} = \frac{z}{2}$

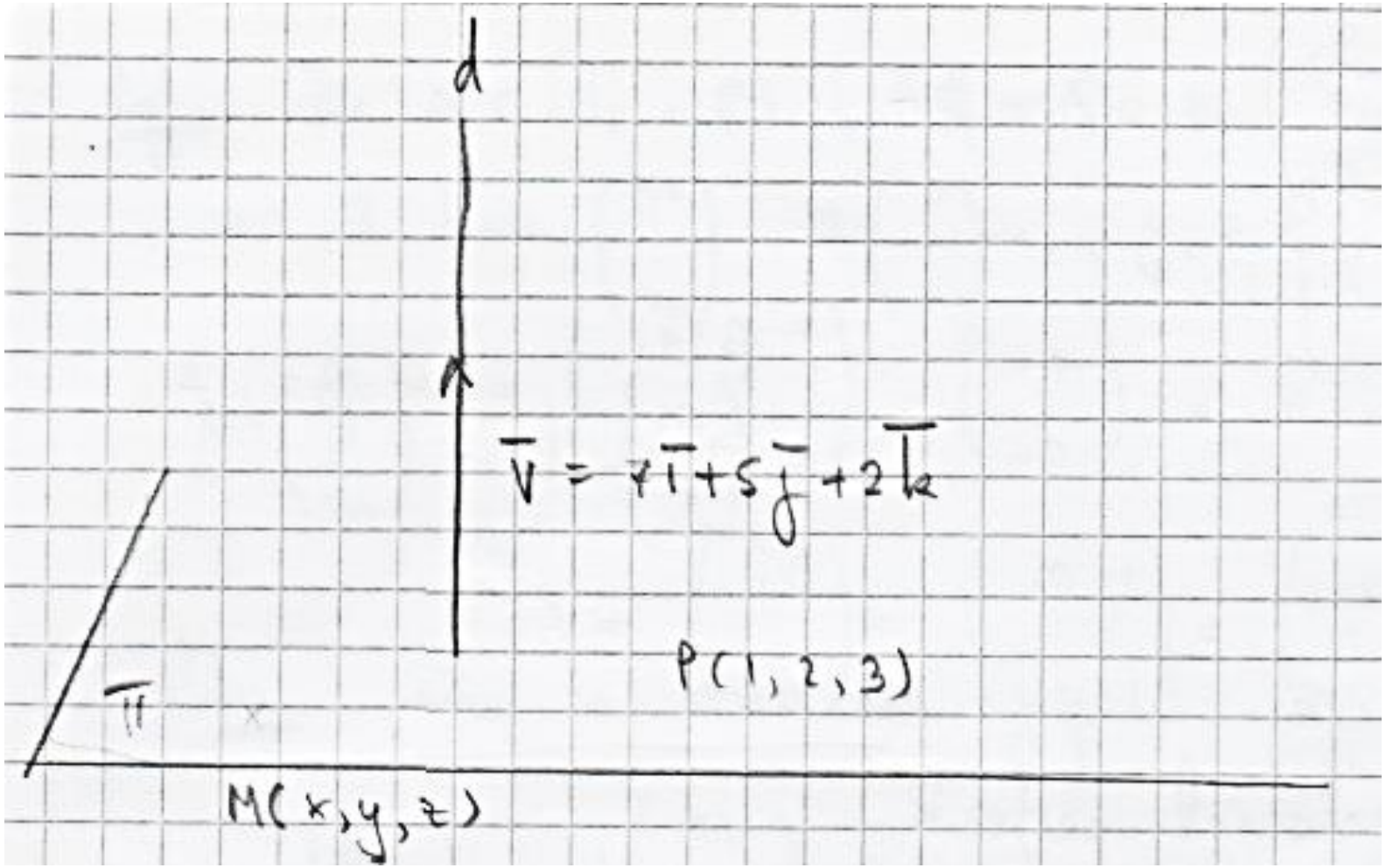
$\bar{v} = 7\bar{i} + 5\bar{j} + 2\bar{k}$

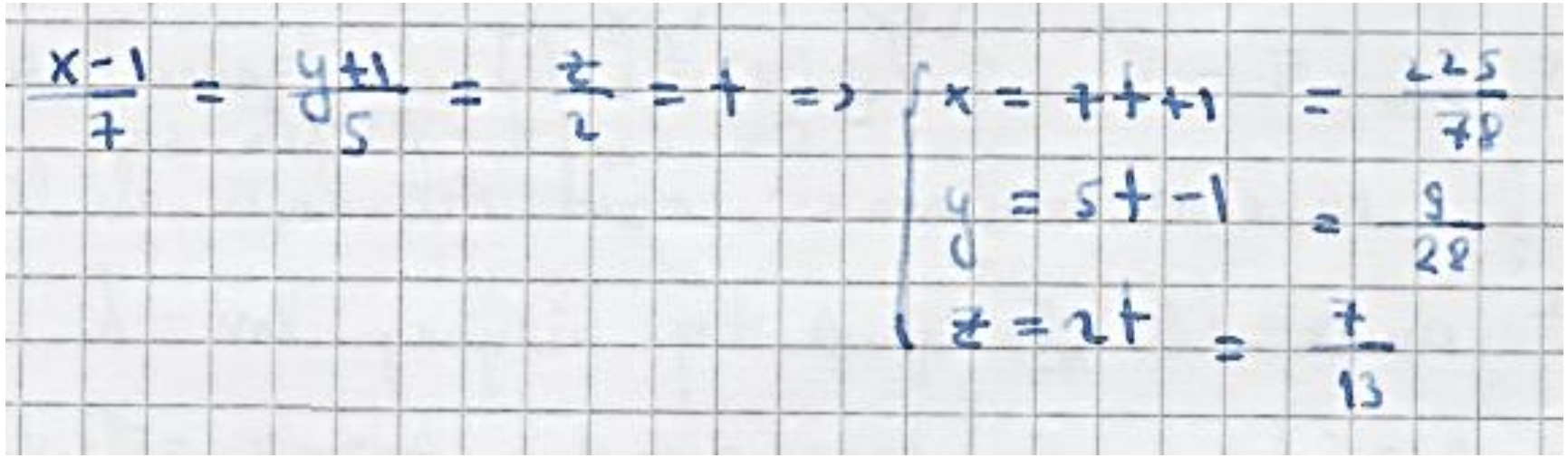


$M \in \Pi \iff \overline{PM} \perp \overline{v} \iff \overline{PM} \cdot \overline{v} = 0$

$\overline{PM} = (x-1)\,\overline{i} + (y+2)\,\overline{j} + (z-3)\,\overline{k}$

$\overline{PM} \cdot \overline{v} = 0 \Rightarrow (x-1) \cdot 1 + (y+2) \cdot 5 + (z-3) \cdot 2 = 0$


d
$\bar{v} = 4\bar{i} + 5\bar{j} + 2\bar{k}$
P(1, 2, 3)
π
M(x, y, z)


$\frac{x-1}{7} = \frac{y+1}{5} = \frac{z}{2} = t \Rightarrow \begin{cases} x = 7t+1 \\ y = 5t-1 \\ z = 2t \end{cases}$
$= \frac{225}{78}$
$= \frac{9}{28}$
$= \frac{7}{13}$

$\Pi: (x-1)\cdot 7 + (y-2)\cdot 5 + (z-3)\cdot 2 = 0$

$7t \cdot 7 + (5t-3)\cdot 5 + (2t-3)\cdot 2 = 0$

$49t + 25t - 15 + 4t - 6 = 0$

$78t - 21 = 0 \Rightarrow t = \frac{21}{78}$

P' - simetricul lui P $x_A = \frac{x_P + x_{P'}}{2} \Rightarrow x_{P'} = 2x_A - x_P$

față de plan

$x_{P'} = \frac{372}{78}$

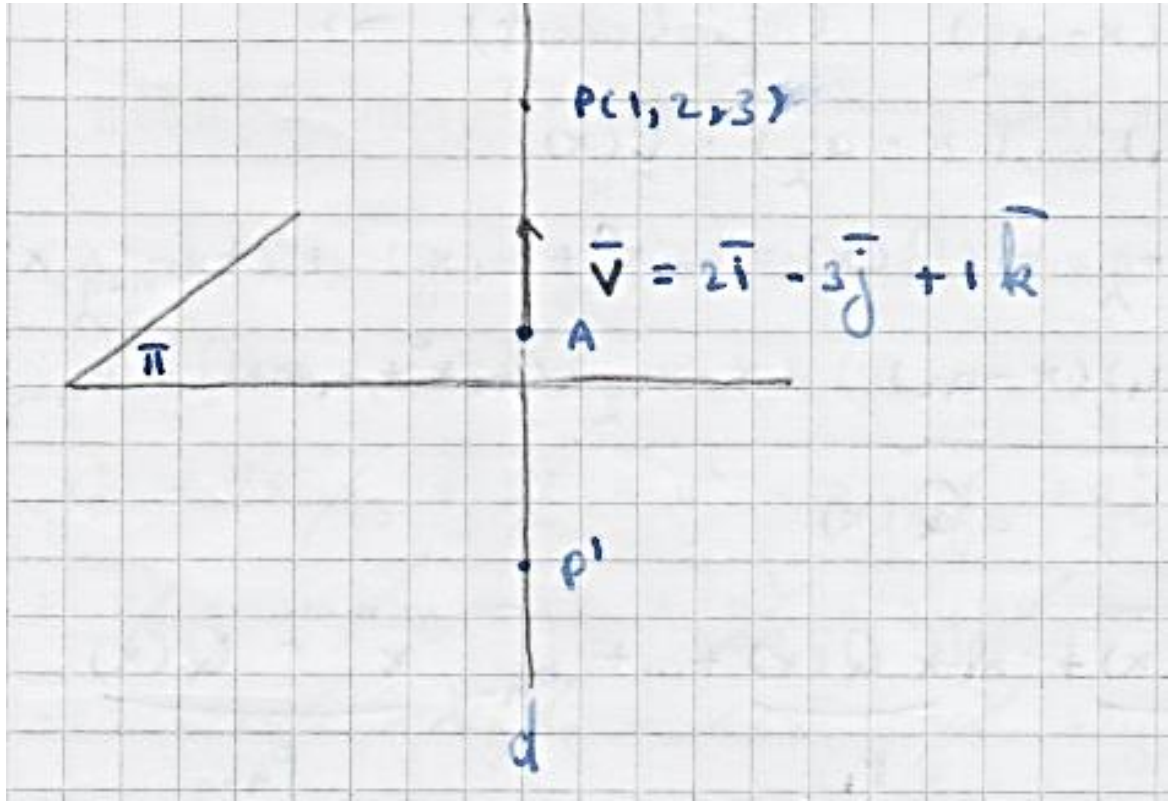
$y_{P'} = \frac{-17}{13}$

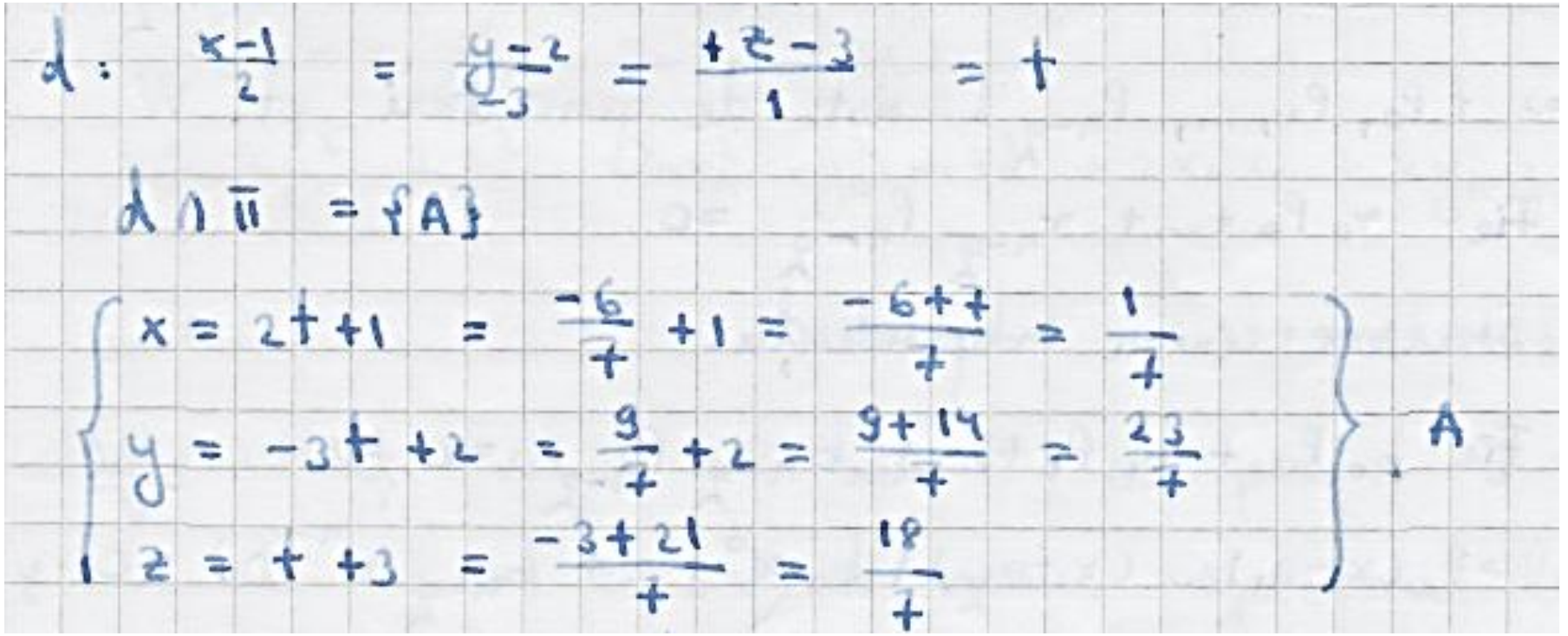
$z_{P'} = \frac{-25}{13}$

$P'\left(\frac{372}{78}, \frac{-17}{13}, \frac{-25}{13}\right)$

Probl.

Să se det. simetricul lui $P(1,2,3)$ în rap. cu planul de ec. generală : $2x - 3y + z + 7 = 0$


P(1, 2, 3)
$\bar{v} = 2\bar{i} - 3\bar{j} + 1\bar{k}$
A
π
P'
d


d: (x-1)/2 = (y-2)/(-3) = (z-3)/1 = t
d ∩ π = {A}
x = 2t + 1 = -6/t + 1 = (-6+t)/t = 1/t
y = -3t + 2 = 9/t + 2 = (9+14)/t = 23/t
z = t + 3 = (-3+21)/t = 18/t
A

$$4t + 2 + 9t - 6 + t + 3 + 7 = 0$$

$$14t + 6 = 0 \Rightarrow t = \frac{-6}{14} = \frac{-3}{7}$$

$$x_A = \frac{x_P + x_P'}{2} \Rightarrow x_P' = 2x_A - x_P = 2 \cdot \frac{1}{7} - 1 = \frac{2-7}{7} = \frac{-5}{7}$$

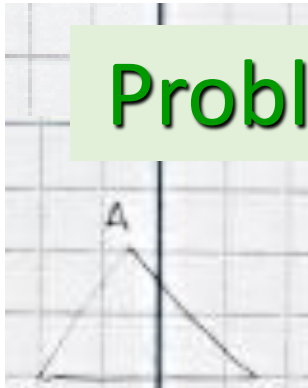
$$y_P' = 2y_A - y_P = 2 \cdot \frac{23}{7} - 2 = \frac{46-14}{7} = \frac{32}{7}$$

$$z_P' = 2z_A - z_P = 2 \cdot \frac{18}{7} - 3 = \frac{36-21}{7} = \frac{15}{7}$$

$$P'\left(\frac{-5}{7}, \frac{32}{7}, \frac{15}{7}\right)$$

Probl.

Să dem. în $\mathcal{V}_3$ T. lui Pitagora generalizată



$$\|\overline{BC}\|^2 = \|\overline{AB}\|^2 + \|\overline{AC}\|^2 - 2\|\overline{AB}\|\cdot\|\overline{AC}\|\cos(\overline{AB},\overline{AC})$$

$$\overline{OA} = x_A\bar{i} + y_A\bar{j} + z_A\bar{k}$$

$$\overline{OB} = x_B\bar{i} + y_B\bar{j} + z_B\bar{k}$$

$$\overline{OC} = x_C\bar{i} + y_C\bar{j} + z_C\bar{k}$$

$$\overline{AB} = \overline{AO} + \overline{OB} = -\overline{OA} + \overline{OB} = (x_B - x_A)\,\bar{i} + (y_B - y_A)\,\bar{j} + (z_B - z_A)\,\bar{k}$$

$$\overline{AC} = \overline{AO} + \overline{OC} = -\overline{OA} + \overline{OC} = (x_C - x_A)\,\bar{i} + (y_C - y_A)\,\bar{j} + (z_C - z_A)\,\bar{k}$$

$$\overline{BC} = \overline{BO} + \overline{OC} = -\overline{OB} + \overline{OC} = (x_C - x_B)\,\bar{i} + (y_C - y_B)\,\bar{j} + (z_C - z_B)\,\bar{k}$$

$$\|\overline{AB}\| = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$$

$$\|\overline{AC}\| = \sqrt{(x_C - x_A)^2 + (y_C - y_A)^2 + (z_C - z_A)^2}$$

$$\|\overline{BC}\| = \sqrt{(x_C - x_B)^2 + (y_C - y_B)^2 + (z_C - z_B)^2}$$

$$\overline{AB} \cdot \overline{AC} = \|\overline{AB}\| \cdot \|\overline{AC}\| \cdot \cos(\overline{AB}, \overline{AC})$$

$$\cos(\overline{AB}, \overline{AC}) = \frac{\overline{AB} \cdot \overline{AC}}{\|\overline{AB}\| \cdot \|\overline{AC}\|}$$